TIN MAI

Nationality: Vietnamese

github.com/creastro
nhutin123@gmail.com
0000-0001-6517-2338

Education

Université Côte d'Azur — Nice, France
Master in Astrophysics and Space Science — Mar. 2025 - Sep. 2026

Università degli studi di Roma Tor Vergata — Roma, Italy
Master in Astrophysics and Space Science — Oct. 2024 - Mar. 2025

University of Science and Technology of Hanoi — Hanoi, Vietnam
Bachelor in Space and Applications — Oct. 2020 - Nov. 2023

Research experience

Observatoire de la Côte d'Azur (OCA, France) — Nice, France
Topic: Modelling the high-resolution spectrum of a directly-imaged exoplanet — Mar. 2026 - Sep. 2026
Advisors: Prof. Vivien Parmentier

- I simulate the high-resolution spectra of the planetary-mass companion VHS 1256 b to investigate its time-variable atmospheric dynamics.

Observatoire de la Côte d'Azur (OCA, France) — Nice, France
Topic: Development of the adaptive optics system installed at C2PU — Dec. 2025 - Feb. 2026
Advisors: Prof. Marcel Carbillet, Dr. Lyu Abe

- I developed a Python code used for assessing the real-time performance of the adaptive optics system at the Calern Observatory.

Astroparticule et Cosmologie Laboratoire (APC, France) — Paris, France
Topic: Measuring shapes of galaxies for weak lensing — Jun. 2023 - Oct. 2023
Advisors: Dr. Axel Guinot, Dr. Cyrille Rosset

- I carried out simulations to examine how accurately the current shape-measurement techniques will perform for future Euclid observations.

Max Planck Institute for Radio Astronomy (MPIfR, Germany) — Remote
Topic: Molecular gas in Carina Keyhole — Oct. 2022 - Mar. 2023
Advisors: Dr. Le Ngoc Tram

- I studied the presence of CO molecules in the vicinity of the intense UV radiation source Eta Carinae, specifically within the region known as Keyhole.

Awards and scholarships

Erasmus Mundus Joint Master (EMJM) scholarships — 2024 - 2026
European Union

Merit Scholarship for Outstanding Students — 2020 - 2023
University of Science and Technology of Hanoi, Vietnam

Silver Medal — Apr. 2023
Vietnam National Mathematics Olympiad

Silver Honour — Jul. 2022
International Astronomy and Astrophysics Competition (IAAC)

Coursework Projects

Note: *Selected final projects completed individually (or in a group but with a large contribution) as part of coursework.*

ELT-PSF[edu] - Understanding images from the Extremely Large Telescope
Course: Astronomical Software Engineering (36 hrs)

- I developed a MATLAB tool that simulates the telescope point-spread function from a user-defined pupil shape (ELT-PSF).

Astronomical observations and data reduction
Course: Experimental Astrophysics (35 hrs)

- I scheduled and carried out observations for exoplanet transits, asteroid light curves, and stellar spectra with a 1-m telescope.

Classifying Gaia sources using Machine Learning techniques
Course: Machine Learning (21 hrs)

- I implemented several machine learning models, including GMM, Random Forest, AdaBoost and SVMs to classify Gaia DR3 sources as stars, quasars or galaxies.

Conferences and Schools

ISYA Vietnam 2026: International School for Young Astronomers
48th International School for Young Astronomers — Oct. 2026

JWST Exosystems proposal preparation workshop for Cycle 6 — June. 2026

TNU/SAGI Summer School in Observational Astronomy — Aug. 2024

SAGI Summer School 2024: Astronomical Techniques — Jul. 2024

Dust Polarimetry and Applications in Astrophysics
The 2023 SAGI Astrophysics Workshop — Nov. 2023

New Frontiers of Astrophysics in the Golden Era of Dust Polarimetry
The 2022 SAGI Astrophysics Workshop — Jul. 2022

Vietnam School on Neutrinos (VSON6) — Jul. 2022

Academic and Outreach

University of Science and Technology of Hanoi — Hanoi, Vietnam
Teaching Assistant - Introduction to Informatics (Bachelor) — Winter 2023
Teaching Assistant - Numerical Methods (Bachelor) — Spring 2024

Vietnam Astrophysics Club (2020-2024) — Hanoi, Vietnam
Content Writer + Editor — Nov. 2023 - Sep. 2024
Event Manager — Mar. 2022 - Nov. 2023